



Real Time Sign Language Recognition

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Introduction

American Sign Language (ASL) is a vital communication tool for the deaf and hard-of-hearing community, using hand gestures, facial expressions, and body postures. Recent advancements in computer vision, particularly MediaPipe's hand-tracking framework, allow precise detection of hand keypoints in 3D space, making gesture recognition more efficient. This project uses MediaPipe for feature extraction and trains multiple machine learning models, such as Neural Networks and MobileVNet, to classify ASL gestures. The goal is to develop a high-accuracy classifier for static ASL gestures, enhancing communication accessibility. The project leverages a Kaggle ASL dataset and evaluates models based on accuracy, precision, and recall. Future enhancements may include real-time recognition and dynamic gestures.

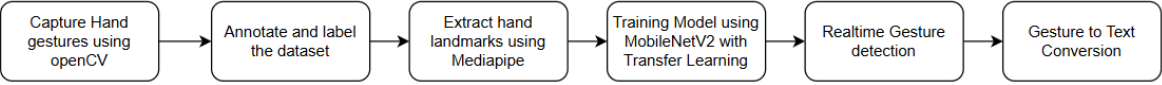
Motivation

This project aims to bridge communication gaps between ASL users and non-users by developing automated recognition systems that provide scalable, on-demand translation. Human interpreters are costly and limited in availability, making automated systems essential for real-time communication in various settings. By focusing on hand keypoints, the project reduces data complexity while maintaining high accuracy, making it suitable for deployment on edge devices. It also explores multiple classifiers to identify the most effective model for practical use. Ultimately, the goal is to enhance accessibility and communication for ASL users.

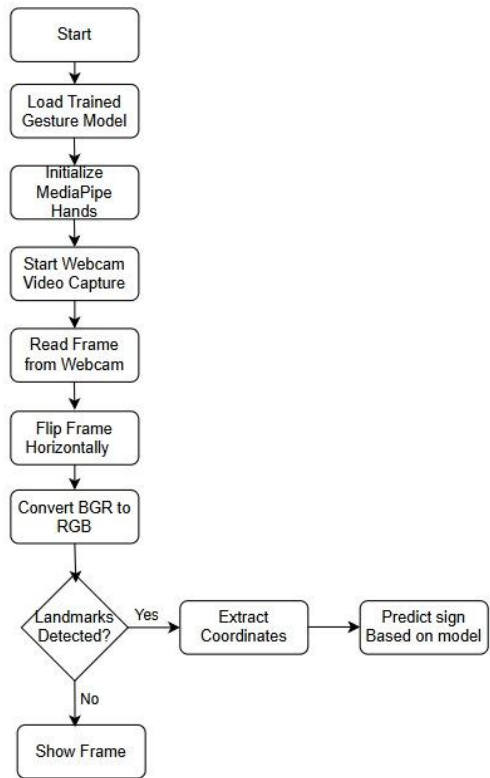
SCOPE of the Project

This project focuses on ASL alphabet recognition using a balanced subset of 1,000 images per class from a Kaggle dataset. Feature extraction is done using MediaPipe's Hands module to derive 21 3D keypoints per image, forming a dataset with 63 features. Three classifiers—Neural Network, Random Forest, and Gradient Boosting—are developed and compared for gesture recognition. Model performance is evaluated through accuracy, precision, recall, F1-score, and confusion matrices. The project also includes visualizations like 3D keypoint plots and performance curves. It is limited to static gestures with one hand per image and excludes dynamic or two-hand signs. Future work could include real-time recognition and dynamic gesture analysis.

Methodology



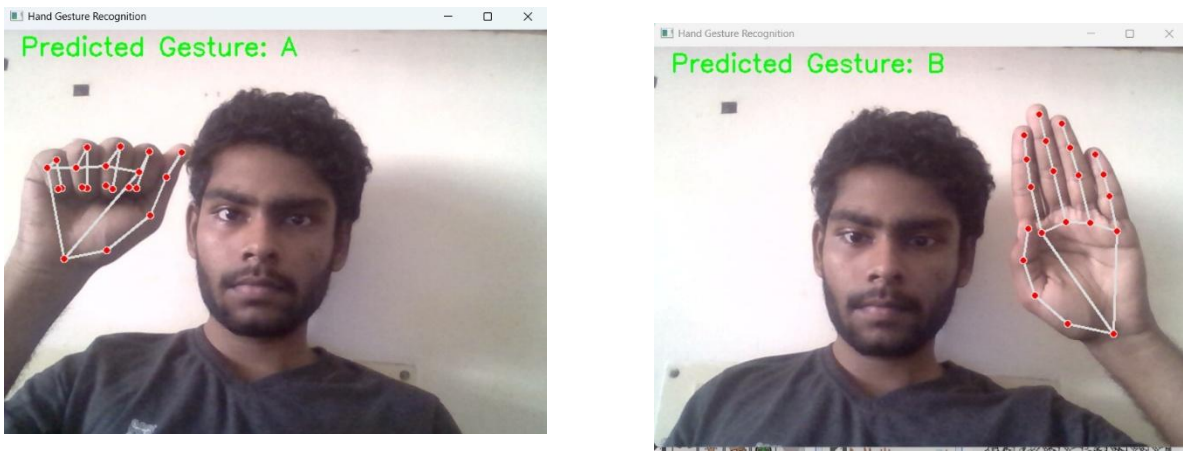
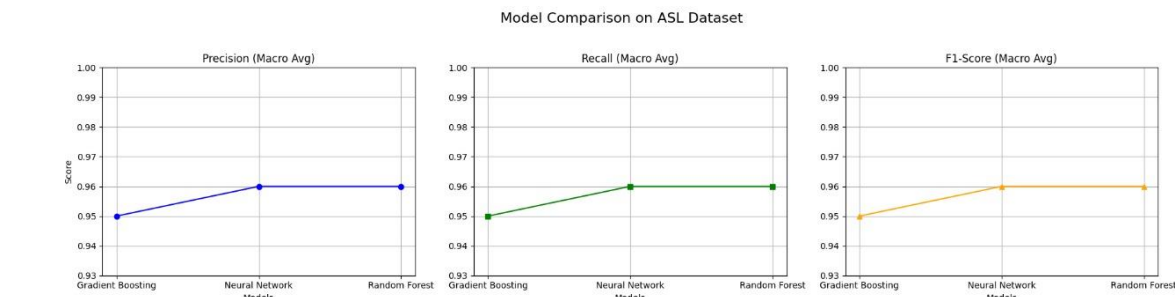
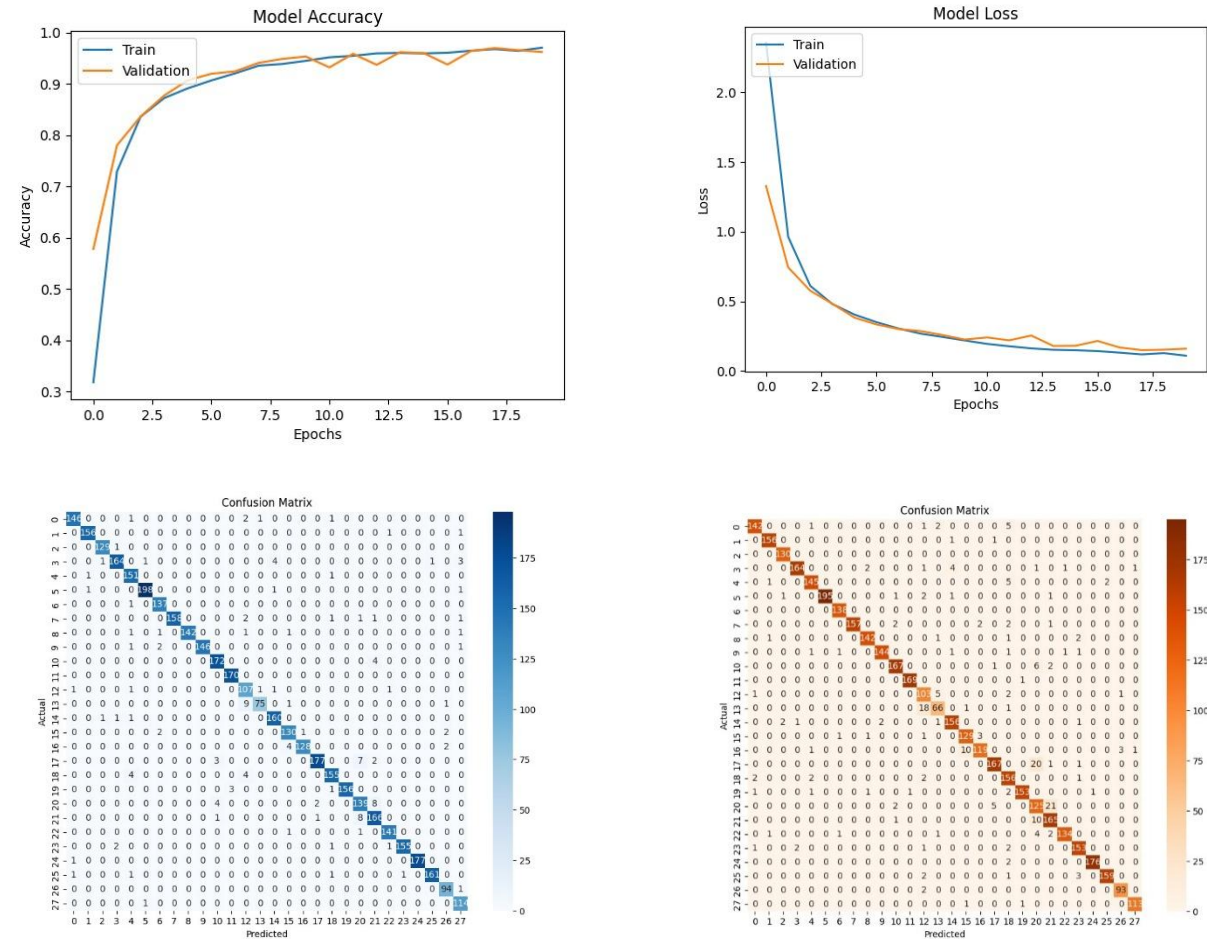
- Step 1: Data collection** – Capture hand gesture images using OpenCV. This involves recording hand movements through a webcam or camera in various positions and angles to build a comprehensive dataset.
- Step 2: Data pre-processing** – Annotate and label the dataset with the corresponding gesture classes. This step involves organizing the collected images and assigning them appropriate gesture labels for supervised learning.
- Step 3: Feature extraction** – Use MediaPipe to extract 21 hand landmarks (keypoints) from each image. These landmarks serve as meaningful features for training the model, significantly reducing the complexity of raw image data.
- Step 4: Model training** – Train a gesture recognition model using MobileNetV2 with transfer learning. This lightweight CNN model adapts to gesture recognition by learning patterns from the extracted landmark features.
- Step 5: Model evaluation and detection** – Implement real-time gesture detection using the trained model. Evaluate its performance in identifying gestures from live camera feeds or video streams.
- Step 6: Output generation** – Convert the detected gestures into corresponding text outputs. This step allows for gesture-to-text translation, enabling communication through recognized hand signs.
- Step 7: Testing and refinement** – Continuously improve the system based on accuracy feedback and user experience. Fine-tune the model or retrain it with more data to increase reliability in diverse lighting and background conditions.



Results

The American Sign Language (ASL) Gesture Recognition System using MediaPipe and multiple machine learning models has been successfully implemented. The system takes input images and extracts hand keypoints using MediaPipe's Hands module, which are then used for gesture classification. The project evaluates three different models: Neural Network, Random Forest, and Gradient Boosting, to identify the most effective model for ASL recognition. After training the models on the ASL alphabet dataset, we used various evaluation metrics, including accuracy, precision, recall, F1-score, and confusion matrices.

Upon assessment, the Random Forest classifier achieved the highest accuracy, outperforming both the Neural Network and Gradient Boosting models. The Neural Network followed closely in terms of accuracy, but the Random Forest demonstrated better overall performance across multiple metrics. The Gradient Boosting model, although effective, had slightly lower performance compared to the other two. The results also included visualizations such as accuracy and loss curves, which further clarified the model's



Conclusion

This project successfully develops an efficient pipeline for ASL alphabet recognition using hand keypoints, leveraging MediaPipe's Hands module to extract 63 3D keypoints per image. By reducing data dimensionality, the system improves training efficiency while focusing on relevant features for gesture classification. Three models—Neural Network, Random Forest, and Gradient Boosting—were evaluated, with Random Forest achieving the highest accuracy. Visualizations such as 3D keypoints and training curves aid model evaluation and debugging. The lightweight design makes the system suitable for real-time, mobile, or edge device applications, contributing to enhanced communication accessibility for the deaf and hard-of-hearing communities..

References

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Saha, S., Saha, M., Das, S., “Deep Learning-based Bangla Sign Language Detection with an Edge Device,” Intelligent Systems with Applications - Elsevier, <https://www.sciencedirect.com/science/article/pii/S2667305323000137>, May 2023